

Haar Phase Equidistribution for a Full-Cycle Open Walsh Gate

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Abstract

We retain the frozen three-symbol open Walsh gate studied at the full register cycle and resolve the phase question left open by its log-modulus law. The two surviving one-site eigenphases have ratio $r = e^{i\delta}$. We derive $2\cos\delta = (\sqrt{3} - \sqrt{111})/6$ and prove that its primitive minimal polynomial is $3x^4 - 19x^2 + 27$. Since this polynomial is nonmonic, $2\cos\delta$ is not an algebraic integer, and r cannot be a root of unity. The full-cycle phase measure is a multiplicity-weighted binomial walk on the circle. Its m th Fourier coefficient is exactly $u_-^{mk}((1+r^m)/2)^k$, so every nonzero fixed mode decays exponentially and the measures converge weakly to Haar measure. Jointly, the centered $\sqrt{k}$ log-modulus fluctuation and phase converge to a product Gaussian–Haar law. A moved-hole control realizes the complementary order-four torsion branch. The theorem concerns only the source-side subunitary scattering gate and gives no self-adjoint or target-spectrum construction.

Keywords: open quantum maps; Walsh dynamics; phase equidistribution; Haar measure; algebraic integers; Fourier coefficients.

中文摘要

本文保持三符号开放 Walsh 门及完整寄存器周期不变，解决此前模长定理遗留的相位问题。两个非零单点特征值的单位相位之比记为 $r = e^{i\delta}$ 。我们精确推出 $2\cos\delta = (\sqrt{3} - \sqrt{111})/6$ ，并证明其本原极小多项式为 $3x^4 - 19x^2 + 27$ 。该多项式非首一，故 $2\cos\delta$ 不是代数整数，从而 r 不可能是单位根。完整周期的相位测度是圆周上的二项乘法游走，其第 m 个傅里叶系数恰为 $u_-^{mk}((1+r^m)/2)^k$ 。每个非零固定模态均指数衰减，因此测度弱收敛到圆周 Haar 测度。进一步，中心化的 $\sqrt{k}$ 对数模涨落与相位联合收敛到高斯 $\square$ Haar 乘积律。移动孔洞对照实现互补的四阶有限子群分支。结论只属于源侧次酉散射门，不构造自伴算子或目标谱。
关键词：开放量子映射；Walsh 动力学；相位等分布；Haar 测度；代数整数；傅里叶系数。

1 Frozen gate and algebraic obstruction

Let $A = F_3^* \text{diag}(1, 0, 1)$ and let B_k be the frozen cyclic register shift. One full cycle is $C_k = B_k^k = A^{\otimes k}$. The two nonzero roots of

$$\lambda^2 - \tau\lambda + q = 0, \quad \tau = \frac{\sqrt{3}}{6} - \frac{i}{2}, \quad q = -\frac{1}{2} - \frac{\sqrt{3}i}{6},$$

are labelled by $|\lambda_+| > |\lambda_-|$. Put $u_{\pm} = \lambda_{\pm}/|\lambda_{\pm}|$ and $r = u_+/u_- = e^{i\delta}$.

Lemma 1. $c = r + r^{-1} = 2\cos\delta = (\sqrt{3} - \sqrt{111})/6$ is not an algebraic integer; in particular r is not a root of unity.

Proof. The exact one-site invariants are $|\tau|^2 = 1/3$, $|\lambda_+|^2 + |\lambda_-|^2 = (1 + \sqrt{37})/6$, and $|\lambda_+\lambda_-| = 1/\sqrt{3}$. Expanding $|\lambda_+ + \lambda_-|^2$ gives the formula for c , and

$$c^2 = \frac{19 - \sqrt{37}}{6}, \quad 3c^4 - 19c^2 + 27 = 0.$$

The polynomial of c^2 has discriminant 37. Also $c \notin \mathbb{Q}(\sqrt{37})$, since otherwise its radical expression would put $\sqrt{3}$ in that distinct quadratic field. Hence the displayed primitive quartic is minimal. Its leading coefficient is 3, proving nonintegrality. If r were a root of unity, $r + r^{-1}$ would be an algebraic integer, a contradiction. $\square$

2 All-length Fourier law

Sampling the nonzero spectrum of C_k with algebraic-multiplicity weight gives

$$\mu_k = 2^{-k} \sum_{j=0}^k \binom{k}{j} \delta_{u_-^k r^j}.$$

Theorem 2. For every $m \in \mathbb{Z}$,

$$\hat{\mu}_k(m) = u_-^{mk} \left(\frac{1 + r^m}{2} \right)^k. \quad (1)$$

Thus μ_k converges weakly to normalized Haar measure on $\mathbb{T}$.

Proof. Equation (1) is the binomial theorem. Lemma 1 gives $r^m \neq 1$ for every $m \neq 0$, whence $|\widehat{\mu}_k(m)| = |\cos(m\delta/2)|^k \rightarrow 0$. Trigonometric-polynomial density completes the proof. $\square$

For $p(z) = \sum_{|m| \leq M} a_m z^m$, the same argument gives the finite-cutoff bound

$$|\mu_k(p) - \int p d\text{Haar}| \leq \sum_{0 < |m| \leq M} |a_m| |\cos(m\delta/2)|^k. \quad (2)$$

3 Joint law and the torsion control

Let $d = \log(|\lambda_+|/|\lambda_-|)$, $\sigma^2 = d^2/4$, and $Y_k = \sqrt{k}(k^{-1} \log |\rho| + \log(3)/4)$. With J_k fair binomial, $Y_k = d(J_k - k/2)/\sqrt{k}$ and the phase is $u_-^k r^{J_k}$. Therefore

$$\mathbb{E}[e^{itY_k} \text{phase}(\rho)^m] \quad (1)$$

$$= u_-^{mk} e^{-itd\sqrt{k}/2} \left(\frac{1 + r^m e^{itd/\sqrt{k}}}{2} \right)^k. \quad (3)$$

For $m = 0$ this tends to $e^{-\sigma^2 t^2/2}$; for $m \neq 0$ its base tends to a number of modulus below one. Hence

$$(Y_k, \text{phase}(\rho)) \implies N(0, \sigma^2) \otimes \text{Haar}, \quad (4)$$

so the two coordinates become asymptotically independent.

More generally, if a binary phase ratio has exact order h , finite Fourier inversion gives convergence in total variation to the uniform measure on the moving coset $u_-^k \langle r \rangle$, with

$$\text{TV} \leq \frac{h-1}{2} \max_{1 \leq m < h} |\cos(\pi m/h)|^k. \quad (5)$$

Moving the frozen hole to $\text{diag}(0, 1, 1)$ gives nonzero roots $-i, -1/\sqrt{3}$, phase ratio i , and the order-four branch $\text{TV} \leq (3/2)(\sqrt{2}/2)^k$. Projector order is a unitary similarity and preserves the non-torsion branch. The closed parent has three phases and is outside this binary statement.

4 Evidence and boundary

The producer freezes 32 binomial/control ledgers and the exact rational recurrence for $r^m + r^{-m}$ through $m = 24$. Independent strict and symbolic reconstructions, byte replay, and hostile mutations test the implementation; the finite receipts do not prove the all- k theorem. The result is source-side and subunitary. It asserts no self-adjoint or antiunitary limit, target divisor or counting law, arithmetic local factor, Euler factor, root number, automorphy, Hilbert–Polya operator, or Route-B authorization. The literal scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.